DCGAN (Deep Convolutional Generative Adversarial Network)

- Goal

Train two convolutional networks:

- Generator (G) learns to create realistic images
- Discriminator (D) learns to distinguish real vs fake images

They compete until generated images look real.

- Math

(Generator):

- Input: random noise $[z \sim N(0, I)$
- Output: fake Image $x\_fake = G(z)$

Uses transposed convolutions to go from:

"noise → low-res features → image"

(Discriminator):

- Input: image x.
- Output: probability image is real $D(x) \in [0,1]$

Uses convolutions to extract spatial features.

- Loss Functions

Discriminator loss

$$L\_D = -[\log D(x\_real) + \log(1 - D(G(z)))]$$

→ wants:

- D (real) → 1
- D (fake) → 0

Generator loss

$$L\_G = -\log D(G(z))$$

→ wants:

- D (fake) → 1 (fool the discriminator)

Training Game

$$\min_G \max_D E[\log D(x)] + E[\log(1 - D(G(z)))]$$

This is a minimax game.

- **Conclusion:**
  - DCGAN replaces dense layers with convolutions
  - Generator upsamples noise → image
  - Discriminator downsamples image → probability
  - Training is unstable but powerful.
  - At balance: $D(\text{real}) \approx D(\text{fake}) \approx 0.5$